Orbit intersections and marked semidirect products

Human Compression Project

Abstract

Orbit intersections give a common construction of flag-transitive geometries for semidirect products, including actions with a nontrivial chamber stabilizer. We prove all parabolic and panel formulas, residual connectivity and the faithful regular specialization. Element orbits and orbital Coxeter presentations supply the diagonal constructions. For signed permutation markings, a separate nested-module argument and an explicit cocycle change of coordinates recover six families, including the odd-rank case left unverified in the source. This edition extends the complete earlier regular theory; it remains a private draft pending source and proof review.

1 Conventions and the two intersection properties

All type sets in this paper are finite. Cosets are left cosets. In a coset geometry $(B, (B_i)_{i\in I})$ write $B_J = \bigcap_{i\in J} B_i$, with $B_\emptyset = B$. A regular hypertope is thin, residually connected and flag-transitive. Automorphisms preserve types. In its faithful standard model, the group acts freely and transitively on chambers. Its distinguished involutions b_i interchange a fixed chamber and its i-adjacent chamber; then

$$B_J = \langle b_i : i \in I \setminus J\rangle, \qquad B_I = 1. \tag{1}$$

We recall why the geometric hypotheses give these assertions. In a thin geometry a type-preserving automorphism fixing a chamber fixes all its adjacent chambers. Residual connectivity implies that the chamber graph, and the chamber graph of every residue, is connected: by induction on rank, chambers containing a given element are connected; a path in the incidence graph connects these families, using chambers containing each consecutive incident pair. The rank-one and rank-two cases start the induction. Thus a chamber stabilizer is trivial. The same argument inside the residue of the base flag of types J proves (1). It also proves that the full type-preserving automorphism group is the displayed group whenever a faithful group already acts transitively on chambers.

For a second regular hypertope $(A, (A_x)_{x\in S})$, put $A(T) = \langle a_x : x \in T\rangle$ for $T \subseteq S$. Formula (1) gives

$$A(T) \cap A(T') = A(T \cap T'). \tag{2}$$

We will use both this intersection property and flag-transitivity. They are different requirements: the intersection property alone is not a general criterion for a nonstring coset system to be a geometry.

Lemma 1 (Coset intersection)**.** *If B is flag-transitive on its coset geometry, every finite pairwise-intersecting family of cosets of standard parabolic subgroups has nonempty total intersection. In particular, for $H, K, J \subseteq I$,*

$$B_H B_J \cap B_K B_J = B_{H\cup K} B_J.$$

Proof. Expand a coset gB_H as $\bigcap_{i\in H} gB_i$. Pairwise intersection of the original cosets implies pairwise intersection of all the resulting maximal-parabolic cosets. Intersecting cosets of the same type are equal. They therefore form a flag, which flag-transitivity takes to a standard flag; its cosets have a common group element. For the last assertion, if $g \in B_H B_J \cap B_K B_J$, the three cosets B_H, B_K, gB_J intersect pairwise. A point of their common intersection belongs to $B_{H\cup K}$ and to gB_J. The reverse inclusion follows from subgroup containment. □

Theorem 2 (Element orbits)**.** *Fix $j \in I$ and $x_0 = B_j \in X = B/B_j$. If B is flag-transitive, then*

$$(B_H x_0) \cap (B_K x_0) = B_{H \cup K} x_0 \qquad (H, K \subseteq I). \tag{3}$$

Thus the complementary notation $O(M) = B_{I \setminus M} x_0$ satisfies $O(M) \cap O(N) = O(M \cap N)$.

Proof. The inverse image of $B_H x_0$ under $B \to B/B_j$ is $B_H B_j$. Apply Lemma 1 and take images. Finally $(I \setminus M) \cup (I \setminus N) = I \setminus (M \cap N)$. □

2 Parabolics with a nontrivial chamber stabilizer

We now allow flag-transitive coset geometries whose acting groups are not faithful. The type sets remain finite and nonempty. In a geometry $\alpha = (A, (A_x)_{x \in S})$ put $A_E = \bigcap_{x \in E} A_x$, with $A_\varnothing = A$, and call A_S the Borel subgroup. It need not be trivial. The residue of the standard flag of types E is the coset geometry of A_E with parabolics $A_{E \cup \{x\}}$, $x \notin E$. To see this identification, take an element incident with that flag; flag transitivity supplies a common representative in A_E. Two such representatives define the same element exactly when their quotient lies in $A_E \cap A_x$. This also identifies incidence by intersecting the corresponding cosets.

For a coset geometry of rank at least two the incidence graph is connected if and only if its maximal parabolics generate the acting group. Indeed, if H is that generated subgroup, paths from the standard flag remain among cosets with representatives in H. Conversely multiplication successively by elements of the parabolics provides paths through incident cosets; when two successive steps have the same type, an intervening different type gives the same conclusion. Thus the components are the translates of this component. Applied to every residue, this proves that residual connectedness is equivalent to

$$A_E = \langle A_{E \cup \{x\}} : x \notin E \rangle \quad \text{if } |S \setminus E| \geq 2. \tag{4}$$

For rank one residual connectedness imposes no condition.

Lemma 3 (Generation of parabolics)**.** *If α is residually connected, then*

$$A_E = \langle A_S,\ A_{S \setminus \{x\}} : x \notin E \rangle.$$

Consequently, for any nonempty family (E_ν) of subsets of S,

$$\langle A_{E_\nu} : \nu \rangle = A_{\bigcap_\nu E_\nu}. \tag{5}$$

Proof. Induct on $|S \setminus E|$. For zero this is $A_S = A_S$; for one the displayed generator is A_E itself. For larger complements, use (4) and the induction hypothesis on each child. Every missing type remains missing in at least one child, so all and only the indicated minimal parabolics occur. For the second assertion, each group on the left lies in the right-hand group. Conversely, every minimal parabolic required to generate the right-hand group occurs in some A_{E_ν}, and A_S lies in every one. The nonempty-family hypothesis is necessary when $A_S \neq 1$. □

The explicit A_S in the first formula must not be dropped at $E = S$. This avoids the empty-generated-subgroup problem in the source's background equivalences.

3 A general orbit construction

Let $\alpha = (A, (A_x)_{x \in S})$ and $\beta = (B, (B_i)_{i \in I})$ be flag-transitive coset geometries. Suppose a specified homomorphism $B \to \mathrm{Aut}(A)$ and an action of B on the indexed set S satisfy $b(A_x) = A_{bx}$. The

indexed action is part of the data, even if different maximal parabolics happen to coincide. Let $\mathcal{K}$ be the set of orbits on S, choose $x_L \in L$ for each $L \in \mathcal{K}$, and assume

$$B_H x_L \cap B_K x_L = B_{H\cup K} x_L \quad (H, K \subseteq I,\ L \in \mathcal{K}). \tag{6}$$

For $T = I \sqcup \mathcal{K}$, define subgroups of $G = A \rtimes B$ by

$$G_i = A_{S\setminus\bigcup_L B_i x_L} \rtimes B_i \quad (i \in I), \qquad G_L = A_L \rtimes B \quad (L \in \mathcal{K}). \tag{7}$$

Every indicated subset of S is invariant under its second factor, so these are indeed subgroups. For $J \subseteq T$ write $J_B = J \cap I$, $J_A = J \cap \mathcal{K}$, and set

$$E(J) = S \setminus \bigcup_{L\in\mathcal{K}\setminus J_A} B_{J_B} x_L.$$

Theorem 4 (Orbit construction with a Borel subgroup)**.** *The coset system in (7) is a flag-transitive geometry. All its standard parabolics, including its Borel subgroup, are*

$$G_J = A_{E(J)} \rtimes B_{J_B}, \qquad G_T = A_S \rtimes B_I. \tag{8}$$

It is finite if and only if both input geometries are finite. It is residually connected if both inputs are residually connected, and firm if both inputs are firm. Its numbers of chambers through panels are

$$[G_{T\setminus\{i\}} : G_T] = [B_{I\setminus\{i\}} : B_I] \qquad (i \in I), \tag{9}$$
$$[G_{T\setminus\{L\}} : G_T] = [A_{S\setminus B_I x_L} : A_S] \qquad (L \in \mathcal{K}). \tag{10}$$

In particular, if both inputs are thin and B_I fixes each chosen x_L, the new geometry is thin. Faithfulness of either input is not needed for this sufficient condition.

Proof. Intersections of subgroups $A_E \rtimes B_H$ are computed coordinate by coordinate. The intersection of their A-parts is obtained by taking the union of their subscripts. Different B-orbits are disjoint, and iteration of (6) says that the intersection of the old supports is the support for B_{J_B}. Intersecting with the orbit parabolics deletes precisely the orbits in J_A. Taking complements proves (8). It uses intersections of maximal parabolics, not an unjustified formula for generated supports.

To prove flag transitivity, project a flag of G-cosets to B. Old types give pairwise-intersecting maximal-parabolic cosets; orbit types project to all of B. Coset intersection from the first section gives a common $b \in B$. After left translation by $(1, b^{-1})$, each coset has a representative $(a, 1)$. Such a coset is the Cartesian product of a coset of a standard parabolic of A and the corresponding subgroup of B. Pairwise incidence therefore gives a pairwise-intersecting family of standard-parabolic A-cosets. Their common representative, again by the coset intersection lemma, realizes the entire flag. Thus every flag is an image of a standard flag and extends to a chamber. This step by itself does not require the orbit identity; that identity is needed for the exact parabolic description and the later conclusions.

Suppose both inputs are residually connected, and fix J with at least two missing types. We prove generation of G_J by the next parabolics. If no orbit type is missing, all their A-parts are A_S; (4) for B gives the required generation. If no old type is missing, choose two distinct missing orbits L, M. The two corresponding next parabolics both contain B_I. Their A-subscripts are $E(J) \cup L$ and $E(J) \cup M$, whose intersection is $E(J)$. Lemma 3 gives the entire A-part.

In the remaining case choose a missing old type i and a missing orbit type L. The parabolic $G_{J\cup\{L\}}$ contains the full B_{J_B}. Write $E' = E(J\cup\{i\})$. The group generated by these two next parabolics therefore contains $A_{bE'}$ for every $b \in B_{J_B}$. In each remaining orbit, $B_{J_B\cup\{i\}} x_M$ is nonempty and contains x_M. Its B_{J_B}-saturation is exactly $B_{J_B} x_M$. Consequently

$$\bigcap_{b\in B_{J_B}} bE' = E(J).$$

There are only finitely many distinct subsets of the finite set S, so Lemma 3 applies directly to these conjugate parabolics. They generate $A_{E(J)}$, proving the remaining case. The residue identification and connectivity criterion already proved now establish residual connectedness in every residue.

Putting $J = T \setminus \{t\}$ in (8) gives the two panel formulas. For an old type the A-parts agree; for an orbit type the B-parts agree. These observations prove the stated indices even when the Borel subgroups are nontrivial. If the inputs are firm, the old indices are at least two. The orbit $B_I x_L$ contains x_L, so $A_{S\setminus\{x_L\}} \le A_{S\setminus B_I x_L}$; its index over A_S is at least two by firmness of α. This proves firmness. If the inputs are thin and $B_I x_L = \{x_L\}$, all the displayed indices are two, proving the sufficient thinness assertion.

Finally the number of chambers is the index of G_T. The elementary coset index formula gives

$$[G : G_T] = [A : A_S][B : B_I],$$

with the equality interpreted as a product of cardinalities. For a finite number of types, a geometry has finitely many elements if and only if it has finitely many chambers: every element extends to a chamber and a chamber is a tuple of one element of each type. The same argument applies to each input. Thus the product is finite exactly when both input geometries are finite. No finiteness of the possibly unfaithful groups A and B was assumed. □

Corollary 5 (Faithful regular inputs)**.** *For faithful regular hypertopes α, β, the construction is a regular hypertope with full type-preserving automorphism group $A \rtimes B$. Writing $A(R) = \langle a_x : x \in R \rangle$, its parabolics are*

$$G_J = A\left(\bigcup_{L \not\subseteq J_A} B_{J_B} x_L\right) \rtimes B_{J_B}. \tag{11}$$

Its distinguished involutions are the b_i and the a_{x_L}.

Proof. The faithful regular model has $A_S = B_I = 1$ and $A_E = A(S \setminus E)$, as proved in the first section. The preceding theorem supplies flag transitivity, residual connectedness and thinness. Its Borel subgroup is trivial, and the panel groups are precisely the cyclic groups on the indicated involutions. Chamber propagation shows that the displayed regular action is the full type-preserving group. □

This recovers the general finite-type twisting theorem and its regular specialization in Piedade–Tranchida [2, Theorem 3.16 and Corollary 3.17]. The exact panel formula also explains why passing to a faithful quotient of B cannot silently preserve an arbitrary action on the types of α: its Borel subgroup must act trivially for that action to descend.

4 Invariant quotients and orbital presentations

Lemma 6 (Quotients)**.** *If $M \le A$ is invariant under B, its normal closure N in A is invariant. The action descends to A/N and permutes the images of the standard maximal parabolics with the same indexed type action. If their coset geometry is a regular hypertope, the corresponding twisting is again a regular hypertope.*

Proof. The normal closure is generated by ama^{-1} with $a \in A, m \in M$. An automorphism from B carries each such generator to another of the same form, and its inverse gives equality. The quotient action and parabolic permutation follow. Orbit identity (6) concerns only the indexed types and is unchanged.

For completeness, the regular-geometry assumption can be interpreted without silently assuming a faithful quotient action. Put $\bar{A} = A/N$, $C_x = A_x N/N$, and $C_E = \bigcap_{x \in E} C_x$, with $C_\varnothing = \bar{A}$. The source's quotient twisting uses intersections of these maximal parabolics:

$$\bar{G}_i = C_{S\setminus\bigcup_L B_i x_L} \rtimes B_i \quad (i \in I), \qquad \bar{G}_L = C_L \rtimes B \quad (L \in \mathcal{K}). \tag{12}$$

These are not defined by substituting the images of generated-support subgroups into (11).

Each image of a_x fixes the base flag with type x deleted. It cannot belong to C_x: otherwise C_x, already containing the other generators, would be all of $\bar{A}$, giving only one element of type x, contrary to thinness. It therefore performs the unique adjacency of that type. Chamber-graph connectivity makes $\bar{A}$ flag-transitive. Its chamber stabilizer $H = C_S$ fixes every chamber by thinness, so is its normal action kernel. It is B-invariant because it is the intersection of the permuted C_x.

Every C_E contains H, and intersections commute with passage to $\bar{A}/H$. In that faithful regular model formula (1) therefore gives

$$C_E/H = \bigcap_{x \in E} (C_x/H) = \langle a_x H : x \in S \setminus E \rangle.$$

Consequently every subgroup in (12) is the full inverse image of the corresponding subgroup from Theorem 5 for $\bar{A}/H$. Apply that theorem. Passage through the normal subgroup $H \rtimes 1$ preserves all cosets and their intersections, so the original quotient twisting has the same regular geometry. For a transitive type orbit $L = S$, its new parabolic is $H \rtimes B$, not necessarily B. The quotient group $\bar{A}$ itself is not asserted to be the faithful automorphism group. □

Proposition 7 (Orbital presentation)**.** *Let B act transitively on a finite set X with base point x_0 and stabilizer P. Give each orbit of unordered distinct pairs a Coxeter label. Let Y be a finite set fixed by B, with fixed labels within Y and one label m_y between y and every element of X. Put $A = W(Y \sqcup X)$. Then $A \rtimes B$ has the presentation consisting of a presentation for B, one involution $a = a_{x_0}$, the involutions c_y for $y \in Y$, and:*

1. *$[a, p] = 1$ for generators p of P, and $[c_y, b] = 1$ for generators b of B;*
2. *the prescribed relations among the c_y, and $(c_y a)^{m_y} = 1$ when $m_y < \infty$;*
3. *$(a\tau a\tau^{-1})^m = 1$ for one representative $\{x_0, \tau x_0\}$ of each unordered pair orbit of finite label m.*

The unordered orbits are indexed by the inverse-paired double cosets $P\tau P \cup P\tau^{-1}P$ with $\tau \notin P$.

Proof. In the proposed group define $a_{bx_0} = bab^{-1}$. The stabilizer relations make this independent of b. Double-coset multiplication changes the representative without changing its pair orbit, and inversion interchanges its endpoints. Conversely, a transporter of two unordered pairs either fixes the selected first endpoint or interchanges the endpoints, giving exactly the two double cosets displayed. Hence conjugates of the chosen relations give every Coxeter relation on X. Conjugating the relations with c_y, which is fixed by B, gives all the mixed Coxeter relations. These elements therefore define a homomorphism from $W(Y \sqcup X) \rtimes B$ onto the proposed group. The reverse map takes the proposed generators to their named elements in the semidirect product; all the relators hold there. Both composites fix the generating set, proving the isomorphism. This is a presentation of the full group, not merely a finite image. □

Lemma 8 (Exact doubled label)**.** *In $A \rtimes B$ let $a = a_x$ and let $b \in B$ be an involution with $bx \neq x$. If $a_x a_{bx}$ has order m (possibly infinite), then ab has order $2m$, with $2\infty = \infty$.*

Proof. We have $(ab)^2 = a_x a_{bx}$. Projection to B sends ab to the nontrivial involution b, so any power of ab equal to 1 has even exponent. For an even exponent $2k$ it is 1 precisely when $(a_x a_{bx})^k = 1$. This proves exactness in both the finite and infinite cases. □

5 Marked permutation modules and changes of coordinates

Let $r \geq 2$, let V be an elementary abelian 2-group, regarded as an $\mathbb{F}_2[S_r]$-module, and let $z \neq 0$ be fixed by the stabilizer of 1 in S_r. Write $s_i = (i\ i+1)$ for $1 \leq i < r$. For each j put $z_j = \pi z$ for any π

taking 1 to j. This is well-defined: two such permutations differ on the right by an element fixing 1. Let V_z be the span of these r vectors. In $V_z \rtimes S_r$ put

$$\rho_0 = (z, 1), \qquad \rho_i = (0, s_i) \quad (1 \le i < r).$$

The multiplication convention is $(v, \pi)(w, \sigma) = (v + \pi w, \pi\sigma)$.

Theorem 9 (A marked module)**.** *The displayed marks form a string C-group of rank r, with generated group $V_z \rtimes S_r$. This assertion requires no linear independence of the orbit vectors $z_1, \dots, z_r$.*

Proof. All marks are nonidentity involutions. The marks ρ_i with $i \ge 1$ generate the indicated copy of S_r; their conjugates of ρ_0 generate V_z. If $i \ge 2$, then s_i fixes 1 and hence z, so ρ_0 commutes with ρ_i. The other string commutations are those of adjacent transpositions.

We prove every subgroup intersection explicitly. For a set of mark indices $D \subseteq \{0, \dots, r-1\}$ let $E = D \setminus \{0\}$ and $P_E = \langle s_i : i \in E \rangle$. This is the product of the symmetric groups on the connected components of the path with retained edges E: path transpositions generate each component and move no point outside it. Its orbit on 1 is an initial segment $[k(E)]$, where $k(E)$ is one plus the length of the initial run $1, 2, \dots$ contained in E. Set

$$W_D = \begin{cases} \langle z_1, \dots, z_{k(E)} \rangle_{\mathbb{F}_2}, & 0 \in D, \\ 0, & 0 \notin D. \end{cases}$$

The group generated by D is exactly $W_D \rtimes P_E$. Indeed, all its vector generators are conjugates of z under P_E, and those conjugates are precisely the vectors in the displayed span. The other path components fix z and do not introduce any further vector.

For two edge sets E, F, a permutation preserving each block of both component partitions preserves each of their pairwise intersections. Those intersections are exactly the components of the path retaining $E \cap F$: two vertices are in the same component precisely when every edge of their unique interval belongs to both sets. Thus $P_E \cap P_F = P_{E \cap F}$. Also $k(E \cap F) = \min\{k(E), k(F)\}$. The vector spaces spanned by initial segments form a nested chain, even when some orbit vectors are dependent. Consequently $W_D \cap W_{D'} = W_{D \cap D'}$. Intersecting inside the Cartesian set $V_z \times S_r$ proves the full C-group property. □

The elementary coordinate change below applies to any group action on an abelian group, not only to permutation modules.

Lemma 10 (Cocycle change of coordinates)**.** *Let Q act on an abelian group V, written additively. Suppose $c : Q \to V$ satisfies*

$$c(gh) = c(g) + gc(h).$$

Then $\Phi_c(v, g) = (v + c(g), g)$ is an automorphism of $V \rtimes Q$, with inverse Φ_{-c}. It fixes V pointwise and carries any marked subgroup intersection identity to the corresponding identity on the image marks. In characteristic two it is an involution.

Proof. The cocycle identity with $g = h = 1$ gives $c(1) = 0$. For two elements,

$$\Phi_c((v, g)(w, h)) = (v + gw + c(g) + gc(h), gh) = \Phi_c(v, g)\Phi_c(w, h).$$

Addition of $-c(g)$ in the same coordinate is an inverse. The last assertions follow from $c(1) = 0$, from bijectivity, and from $c(g) + c(g) = 0$ in characteristic two. In particular an automorphism commutes with taking generated subgroups and their intersections. □

Proposition 11 (Three pairs of signed markings)**.** *Take $V = \mathbb{F}_2^r$ with its natural permutation action. Write $e_1, \dots, e_r$ for its basis and $\lambda = e_1 + \cdots + e_r$. For any of the three seeds*

$$z = \lambda, \qquad z = e_1, \qquad z = \lambda + e_1,$$

both of the following lists are string C-group markings:

$$(z,1),(0,s_1),\dots,(0,s_{r-1}),\qquad (z,1),(\lambda+e_1+e_2,s_1),\dots,(\lambda+e_{r-1}+e_r,s_{r-1}). \tag{*}$$

The common abstract groups for these three pairs are respectively $C_2 \times S_r$, $C_2 \wr S_r$, *and*

$$\begin{cases} C_2 \wr S_r, & r \text{ even}, \\ V^{\mathrm{even}} \rtimes S_r, & r \text{ odd}, \end{cases} \qquad V^{\mathrm{even}} = \{v : \sum_j v_j = 0\}.$$

In the last line $r \geq 3$ *when odd. The statement includes the odd-rank second list for the third seed.*

Proof. Each seed is nonzero and fixed by the stabilizer of 1, so the first list is covered by Theorem 9. To identify its vector span, only the third seed needs explanation. Its orbit vectors are $\lambda + e_j$. A relation with coefficients b_j says $b = (\sum_j b_j)\lambda$. For even r this forces $b = 0$; for odd r the only nonzero relation is $b = \lambda$. In the odd case every orbit vector has even coordinate sum, so their $(r-1)$-dimensional span is exactly V^{even}.

Let $a \in V$ have alternating coordinates, so that $a_i + a_{i+1} = 1$. Let $\epsilon : S_r \to \mathbb{F}_2$ be the sign homomorphism. Define

$$c(\pi) = \epsilon(\pi)\lambda + a + \pi a.$$

The first summand is a cocycle because λ is fixed and ϵ is additive. The other two are a coboundary: $a + \pi\sigma a = (a + \pi a) + \pi(a + \sigma a)$. For s_i the result is $c(s_i) = \lambda + e_i + e_{i+1}$. Lemma 10, in the full ambient $V \rtimes S_r$, therefore carries the first list in (*) to the second, and fixes its first mark. It need not preserve the first list's named subgroup as a subset; it is an isomorphism onto the subgroup generated by the second list. This distinction matters for odd r, where the image copy of S_r may have odd vector coordinates. It does not affect the intersection property or the asserted abstract group. □

Source correspondence and attribution. The ordinary lists in Proposition 11, in the order of its three seeds, are graph (1) of Table 9 and graphs (13), (14) of the cited Fernandes–Piedade catalogue. The changed-coordinate lists are graph (2) of Table 9 and graphs (15), (16), respectively. In the bound arXiv:2407.16003v2 edition the latter four diagrams are in Table 6, physical page 25; the target paper variously calls that table 5 or 6. The catalogue's Proposition 4.7 and Table 6 concern $r = n/2 \geq 7$; the algebraic marking theorem proved here holds for every $r \geq 2$. Here a vector e_j flips the two points in block j, and s_i interchanges adjacent blocks without flips. Thus the vector $\lambda + e_i + e_{i+1}$ means precisely the vertical i-edges on every block except i and $i+1$, as shown in those diagrams.

The known family statements are Piedade–Tranchida, Proposition 5.1. The odd-rank graph (16) is expressly left unverified by their method in the paragraph immediately before that proposition. The formula above proves that case as well, subject to independent source and proof review. This is not a claim that no previous work settles it. Cocycle changes of splitting are standard group cohomology; the contribution here is their explicit use to identify the paired markings, together with the nested-orbit-span proof which tolerates the odd-rank relation. No source-specific theorem is presumed in either proof.

6 Attribution and retained facts

The constructions and target results are due to Piedade and Tranchida [1,2]. The orbital presentation is the same argument used for the matching extensions in our earlier common paper, with the fixed vertices included explicitly. The contribution of this proof edition is a common general semidirect theorem, exact panel formulas and explicit adapters, not a priority claim for generalized twisting. The marked-module and cocycle arguments expose the common proof of the signed families; their application to the odd-rank case is stated separately from known recoveries. The classical Coxeter facts used

by the application are that standard parabolic subgroups embed with their indicated Coxeter presentations and that the standard coset geometry is a regular hypertope. The usual string C-group–regular-polytope correspondence and the named classical twisting constructions are retained as stated in the application. The companion Lean file verifies only the declared coset, support, and order identities; it is not an end-to-end geometry formalization.

References

[1] C. A. Piedade and P. Tranchida, *Coset geometries acting on their elements: The j-diagonals twisting*, arXiv:2608.23223v1, 2026.

[2] C. A. Piedade and P. Tranchida, *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, 2026, Definition 3.10 and Corollary 3.17.

[3] M. E. Fernandes and C. A. Piedade, *Regular polytopes of rank $n/2$ for transitive groups of degree n*, arXiv:2407.16003v2, 2025; J. Algebra 689 (2026), 256–296. The diagram correspondences use Tables 6 and 9 of the bound arXiv edition.